FAQ: INSTALLATION EFSA PIPELINE RELEASED ON 2026-05-22

This document provides tips for addressing potential issues that may arise during the local installation of the EFSA One Health analytical pipeline. The questions and solutions presented here are based on our own installation experience and may not apply to every installation.

Pipeline:
https://dev.azure.com/efsa-devops/EFSA/_git/efsa.wgs.onehealth
Docker images:
https://efsa-devops@dev.azure.com/efsa-devops/EFSA/_git/efsa.wgs.onehealth.images

NOTE: Compatible only with Nextflow versions ≤ 24.10.

Q1. What if `build.sh` fails to build some Docker images?

If you run the *build.sh* script provided in the Docker images repository and encounter errors while building some images, the first step is to check which images have been successfully installed (`docker image ls`). Table 1 lists the Docker images and their expected tags for the EFSA analytical pipeline. Based on our experience, you may encounter four possible situations:

1. The image is correctly built and has the expected tag. No action is required.
2. The image is correctly built but has the wrong tag. Simply assign the expected tag (e.g. `docker tag local/efsa.supporting.shovill:0.1 local/efsa.supporting.shovill:0.1.0`).
3. The image was not built, but its folder contains a Dockerfile. Navigate to the image folder and build the image directly* (e.g. `docker build -t efsa.supporting.mlst:0.3.0 .`).
4. The image is not available in the efsa.wgs.onehealth.images repository (e.g. chewBBACA). Pull the required version directly from Docker Hub (e.g. `docker pull ummidock/chewbbaca:v3.3.10`).

*Exception: For the ConFindr image, you may also need to create a confindr_db/ directory (`mkdir confindr_db`) in the same directory as the Dockerfile before building the image (`docker build -t efsa.supporting.confindr:0.3.0 .`).

Q2. Where should I save my cgMLST schemas? Are there any restrictions on schema names?

You can save your cgMLST schemas in any place of your system. However, you need to make sure that your schema directory is called **schema_dir**, and that it contains the following directories with the schemas:

- *Escherichia coli*: ecoli_INNUENDO_wgMLST
- *Salmonella enterica*: senterica_INNUENDO_cgMLST
- *Listeria monocytogenes*: lmonocytogenes_Pasteur_cgMLST
- *Campylobacter jejuni**: CAMPISO-PubMLST-8_jejuni*
- *Campylobacter coli**: CAMPISO-PubMLST-8_coli*

Do not forget to indicate the PATH/TO/schema_dir in:
/mnt/extra_disk_04_10TB/EFSA_pipeline_20260522/efsa.wgs.onehealth/configs/params.config

* The cgMLST schema provided with the EFSA One Health analytical pipeline for *C. jejuni* and *C. coli* is the same and is named CAMPISO-PubMLST-8 by default. However, to avoid potential incompatibilities when analyzing both species simultaneously (i.e., by running two independent commands that use the same schema), **we duplicated the schema and renamed each copy as follows: *C. jejuni* - CAMPISO-PubMLST-8_jejuni; *C. coli* - CAMPISO-PubMLST-8_coli**. For this workaround to function correctly, you must also update the schema name on lines 97 and 102 of *modules/allele_calling/allele_calling.nf* using a text/code editor.

Q3. I downloaded my schemas from chewie-NS after June 2025. Do I need to pay attention to loci nomenclature?

In the Summer of 2025, the loci nomenclature of the cg/wgMLST schemas available in chewie-NS had to be modified, due to technical reasons. However, the EFSA analytical pipeline still relies on the previous loci nomenclature system. Therefore, if you downloaded your schema in the Summer of 2025 or afterwards and:

i. intend to submit the allele matrix to EFSA One Health WGS system, you will probably need to convert the loci names to the loci nomenclature used by EFSA. To do so, you can use the information provided by the chewie-NS team. To facilitate, **if you run the *efsa_wgs_onehealth_facilitator.py* script, you can download the *_loci_IDs_mapping.tsv file provided by the chewie-NS team for your species of interest and, then, provide it as**

input to the '--locus-conversion' argument. The *efsa_wgs_onehealth_facilitator.py* will automatically convert the loci names for you.

ii. and are running the EFSA analytical pipeline for *S. enterica* or *E. coli*, you will probably need to replace the list with the cgMLST loci that is provided with the EFSA workflow (see directory data/chewbbaca_filters/) by a list with the new loci names. To do so, please **download the list provided in ReporTree's GitHub and replace the original cgMLST loci list available in data/chewbbaca_filters/. Make sure that you just change the loci names! The file name should be maintained, as the EFSA pipeline will look for it!**

Location of the file for *S. enterica*: data/chewbbaca_filters/cgMLST_salmonella.txt
Location of the file for *E. coli*: data/chewbbaca_filters/cgMLST_ecoli.txt

Q4. Do I have to download the mlst databases? Where can I find them?

Yes, you need to download your own mlst database to be able to run the EFSA analytical pipeline. The mlst schemas of *E. coli*, *S. enterica* and *Campylobacter* spp. are hosted in PubMLST (https://pubmlst.org/). The schema of *Listeria monocytogenes* is hosted in BIGSdb-Pasteur (https://bigsdb.pasteur.fr/). The workflow to get the schemas from each of these databases is very similar:

1. Create an account in the platform
2. Enter your account and register your access to the databases of the species of interest
3. Get an API key for OAuth authentication
4. Download the schemas in the command line using your Client id and Client secret

While in PubMLST you can obtain your key through the PubMLST website, in BIGSdb you will need to contact the team and justify your request (follow these instructions: https://bigsdb.pasteur.fr/requesting-api-key/).

Q5. I have my API key. How do I download the schema(s)? Is there a specific folder structure to store the mlst schemas?

To download the mlst schemas, you can follow the instructions provided at: https://github.com/himal2007/mlst/tree/updatedb_fix/update-fix#readme.

Q6. Which config files do I need to modify?

The EFSA One Health analytical pipeline includes multiple configuration files in the folder configs/, which can be edited according to the specificities of your system:

- *params.config*: [editing is mandatory] this file is where you should indicate the full path to the different databases.

  Example:

```
FILESET_MNT= '/path/to/efsa.wgs.onehealth/schema_dir'
mlst_db = '/path/to/efsa.wgs.onehealth/mlst'
mash_sketches = '/path/to/efsa.wgs.onehealth/mash_db/refseq.genomes.k21s1000.msh'
mash_gram = '/path/to/efsa.wgs.onehealth/mash_db/gram_stain.txt'
```

- *nextflow.config*: [editing is optional] this file is where you can adjust the memory used by different processes.

- *params_listeria-monocytogenes.config*: [editing is optional] this file is where you can adjust the parameters for *Listeria monocytogenes*. If you intend to run the EFSA pipeline as it is, you should not touch this file.

- *params_salmonella-enterica.config*: [editing is optional] this file is where you can adjust the parameters for *Salmonella enterica*. If you intend to run the EFSA pipeline as it is, you should not touch this file.

- *params_escherichia-coli.config*: [editing is optional] this file is where you can adjust the parameters for *Escherichia coli*. If you intend to run the EFSA pipeline as it is, you should not touch this file.

- *params_campylobacter-coli.config*: [editing is optional] this file is where you can adjust the parameters for *Campylobacter coli*. If you intend to run the EFSA pipeline as it is, you should not touch any of the parameters of this file. However, if you placed the cgMLST schema in a directory different from data/Campylobacter_schema/CAMPISO-PubMLST-8/, you should modify it, as the original version of this file does not assume the configuration provided in params.config.

- *params_campylobacter-jejuni.config*: [editing is optional] this file is where you can adjust the parameters for *Campylobacter jejuni*. If you intend to run the EFSA pipeline as it is, you

should not touch any of the parameters of this file. However, if you placed the cgMLST schema in a directory different from data/Campylobacter_schema/CAMPISO-PubMLST-8/, you should modify it, as the original version of this file does not assume the configuration provided in params.config.

Tables

Table 1. List of the images and respective tags expected by EFSA analytical pipeline, as well as the command line you should run if build.sh fails.

Software	Expected Tag	Command line
chewbbaca	ummidock/chewbbaca:v3.3.10	`docker pull ummidock/chewbbaca:v3.3.10`
shovill	local/efsa.supporting.shovill_se:0.1.0	`docker build -t local/efsa.supporting.shovill_se:0.1.0 .`
shovill	local/efsa.supporting.shovill:0.1.0	`docker build -t local/efsa.supporting.shovill:0.1.0 .`
fastp	local/efsa.supporting.fastp:0.3.0	`docker build -t local/efsa.supporting.fastp:0.3.0 .`
mash	local/efsa.supporting.mash:0.7.1	`docker build -t local/efsa.supporting.mash:0.7.1 .`
seqsero2	local/efsa.supporting.seqsero2:0.3.0	`docker build -t local/efsa.supporting.seqsero2:0.3.0 .`
confindr	local/efsa.supporting.confindr:0.3.0	`mkdir confindr_db; docker build -t local/efsa.supporting.confindr:0.3.0 .`
	local/efsa.supporting.sv_extraction:0.16.1	`docker build -t local/efsa.supporting.sv_extraction:0.16.1 .`
mlst	local/efsa.supporting.mlst:0.3.0	`docker build -t local/efsa.supporting.mlst:0.3.0 .`
resfinder	local/efsa.supporting.resfinder:0.5.0	`docker build -t local/efsa.supporting.resfinder:0.5.0 .`
seq_typing	local/efsa.supporting.seq_typing:3.0.1	`docker build -t local/efsa.supporting.seq_typing:3.0.1 .`
checkm	biocontainers/checkm-genome:1.1.3--py_1	`docker pull quay.io/biocontainers/checkm-genome:1.1.3--py_1`